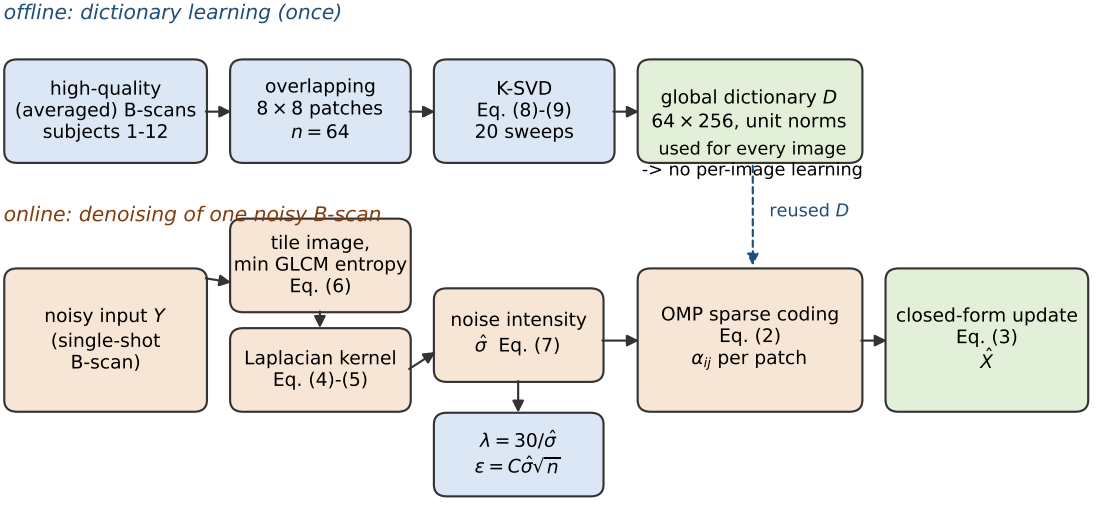
offline: dictionary learning (once)
high-quality (averaged) B-scans subjects 1-12
overlapping 8 × 8 patches n = 64
K-SVD Eq. (8)-(9) 20 sweeps
global dictionary D 64 × 256, unit norms
used for every image
-> no per-image learning
reused D
online: denoising of one noisy B-scan
noisy input Y (single-shot B-scan)
tile image, min GLCM entropy Eq. (6)
Laplacian kernel Eq. (4)-(5)
noise intensity σ̂ Eq. (7)
λ = 30/σ̂
ε = Cσ̂√n
OMP sparse coding Eq. (2) α_ij per patch
closed-form update Eq. (3) X̂